

FERDER FYR, Norway

WMO: 014820

Lat: 59.027N

Lon: 10.524E

Elev: 8

StdP: 101.23

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-8.4	-6.6	-12.0	1.3	-5.9	-10.1	1.6	-4.4	17.4	2.5	16.4	1.9	7.2	30	0.777

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.4	23.1	18.9	22.0	18.2	20.9	17.4	19.7	22.3	18.8	21.4	17.9	20.5	4.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.6	13.4	21.6	17.6	12.6	20.9	16.6	11.9	19.9	56.2	22.5	53.3	21.4	50.5	20.6	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.7	15.2	13.9	-9.3	24.8	2.6	1.7	-11.2	26.0	-12.7	27.1	-14.1	28.0	-16.0	29.3		
(5)			-9.8	20.3	2.3	1.3	-11.5	21.2	-12.9	22.0	-14.2	22.7	-15.9	23.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.6	0.8	0.4	2.1	6.1	11.0	15.0	18.0	17.7	14.2	9.4	5.2	2.4
	DBStd	6.91	3.61	3.61	2.73	2.42	2.79	2.13	2.15	2.14	2.10	2.67	3.31	4.01
	HDD10.0	1361	287	270	244	119	19	0	0	0	1	42	144	235
	HDD18.3	3606	545	503	502	366	227	103	34	39	126	276	393	493
	CDD10.0	841	0	0	0	3	52	151	247	238	125	25	2	0
	CDD18.3	43	0	0	0	0	1	3	22	18	0	0	0	0
	CDH23.3	28	0	0	0	0	1	2	21	6	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	7.2	8.1	7.6	6.9	6.1	6.3	6.1	6.5	7.4	8.0	8.6	8.1	
Precipitation	PrecAvg	802	71	45	39	55	45	64	61	78	78	100	93	68
	PrecMax	1233	138	121	135	117	79	130	129	152	150	234	345	124
	PrecMin	605	10	18	0	10	2	4	8	25	29	32	18	11
	PrecStd	162	34	28	33	29	25	30	28	37	36	49	77	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.1	7.2	8.0	14.1	21.2	22.7	25.4	24.1	19.7	15.2	12.6	9.9
		MCWB	6.5	4.9	5.1	10.1	15.8	16.9	19.4	19.3	16.0	12.9	11.0	8.5
	2%	DB	7.0	6.1	6.9	12.1	18.0	20.7	23.6	22.9	18.4	14.3	11.4	9.1
		MCWB	5.5	4.6	4.4	9.0	13.7	15.7	19.1	19.1	15.5	12.4	10.0	7.8
	5%	DB	6.2	5.4	6.0	10.7	16.1	19.2	22.2	21.9	17.7	13.7	10.6	8.4
		MCWB	4.8	4.3	4.2	8.2	12.6	15.0	18.4	18.5	15.0	11.9	9.1	7.0
	10%	DB	5.3	4.8	5.4	9.4	14.7	18.1	21.1	20.9	17.0	13.1	9.8	7.5
		MCWB	4.1	3.8	3.9	7.2	11.8	14.4	17.6	17.7	14.6	11.4	8.2	6.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.0	5.7	5.9	10.8	16.4	18.3	20.6	20.7	17.5	13.6	11.6	9.0
		MCDB	7.9	6.3	7.1	13.0	20.3	21.7	23.8	22.8	19.1	14.5	12.3	9.7
	2%	WB	5.9	5.0	5.2	9.5	14.5	16.7	19.7	19.9	16.4	13.0	10.3	8.1
		MCDB	6.7	5.7	6.2	11.7	17.2	19.5	22.7	22.2	17.8	13.9	11.2	9.0
	5%	WB	5.1	4.5	4.7	8.5	13.2	15.8	19.0	19.0	15.7	12.4	9.3	7.2
		MCDB	5.9	5.3	5.7	10.3	15.5	18.5	21.8	21.4	17.2	13.4	10.4	8.1
	10%	WB	4.3	3.9	4.1	7.5	12.1	15.0	18.1	18.1	15.0	11.7	8.4	6.4
		MCDB	5.2	4.8	5.1	9.1	14.2	17.4	20.7	20.6	16.7	12.9	9.6	7.4
Mean Daily Temperature Range	5% DB	MDBR	2.9	2.7	2.9	3.4	3.7	3.6	3.4	3.2	2.9	2.8	2.7	2.9
		MCDBR	3.0	2.5	3.3	4.8	5.3	4.9	4.6	4.0	3.2	2.7	3.0	2.6
		MCWBR	3.3	2.4	2.5	3.1	3.2	3.4	2.9	2.8	2.9	2.9	3.6	3.0
	5% WB	MCDBR	2.9	2.3	2.9	4.2	4.6	4.2	4.2	3.6	2.6	2.3	2.7	2.6
		MCWBR	3.2	2.4	2.4	2.9	3.0	3.1	2.8	2.9	2.8	2.8	3.5	3.0
Clear-Sky Solar Irradiance	taub		0.289	0.324	0.341	0.380	0.373	0.360	0.382	0.379	0.349	0.337	0.304	0.265
	taud		2.246	2.326	2.370	2.315	2.413	2.474	2.433	2.436	2.503	2.428	2.290	2.170
	Ebn at Noon		524	694	798	822	858	876	844	819	789	674	498	411
	Edh at Noon		44	68	86	107	104	100	102	95	76	61	42	33
All-Sky Solar Radiation	RadAvg		0.32	0.89	2.39	3.88	5.19	5.90	5.33	4.18	2.78	1.24	0.42	0.21
	RadStd		0.06	0.17	0.24	0.52	0.49	0.37	0.59	0.28	0.29	0.20	0.10	0.02

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.38	N/A	+1.21	N/A	N/A	-0.94	N/A	N/A	N/A	N/A
(47)	Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page